

OSBAMS Rev.2 controller PCB — factory functional test (optional)

Do NOT connect a battery during factory PCB testing. Use only the bench supplies named below. This board has no MCU, no USB and no ADC of its own, so the test list is adapted: steps that need the plug-in Nucleo/INA228 modules are marked as customer-side. Expected values are design estimates, not measured data.

Preconditions: D1/D2/D3 polarity issue (review item R1) resolved and the board rebuilt; external modules NOT fitted unless stated.

#	Step	Method	Expected / pass criterion
1	Visual inspection	Compare with the Assembly Drawing: C1 +, D1–D3 band, Q1/U1 orientation, headers, no solder bridges, no missing parts.	All parts present and oriented per the corrected drawing.
2	Short-circuit check (power off)	Ohmmeter: +12V–GND, 12V_IN–GND, +3V3–GND, COIL_V–COIL_SW, GATE–GND.	No dead shorts (all > 100 ohm; GATE–GND ≈ 10 kΩ via R2).
3	Apply logic power	Bench 3.3 V, current limit 50 mA, to J5 pad 1 (+3V3) and pad 2 (GND). No Nucleo fitted.	Supply draws < 10 mA.
4	Verify 3.3 V rail	Voltmeter on J4.1, J6.1, U1.5, C3.	3.3 V ± 5 % at each point.
5	Program STM32	N/A at the fab — the STM32 is on the Nucleo module (ST-LINK on that module).	Customer-side.
6	Confirm USB/UART	No USB on this board. Continuity J5.6–J6.3 (UART_TX) and J5.7–J6.4 (UART_RX).	Continuity < 1 ohm; no crosstalk between the two.
7	Confirm INA228 communication	Customer-side with the INA228 breakout fitted. At the fab: continuity J4.3–J5.4 (SCL), J4.4–J5.3 (SDA).	Continuity OK; I2C ACK at 0x40 customer-side.
8	Confirm ADC channel	N/A — this PCB has no ADC channel.	Not applicable.
9	Confirm temperature interface	3.3 V on, read TC74 over I2C with a customer adapter at address 0x4D (or check SDA/SCL continuity J5↔U1).	ACK at 0x4D customer-side; continuity J5.3–U1.2, J5.4–U1.4.
10	Confirm relay-driver output	Bench 12 V (current limit 0.5 A) on J1 (pad 1 = +, pad 2 = GND). 24 ohm / 6 W resistor on J2 as a coil stand-in, J3 shorted with a wire. Drive J5.5 to 3.3 V through 1 kΩ, then release.	+12V ≈ 11.4–11.8 V (Schottky drop); COIL_SW ≈ 0.1 V and the load draws ≈ 0.5 A with the gate high; COIL_SW ≈ +12 V and no current with the gate low. Open J3: no current in either state.
11	Reverse-polarity check	Bench 12 V with reversed polarity, current limit 0.2 A.	No current (D3 blocks).

Record: serial/lot, date, operator, measured values and pass/fail for each step. Any failed step = hold the board.